

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.
11. Norman S. Nise, *Control Systems Engineering*, 7th edition, Wiley, 2015.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 34****Topics covered :**

1. Inverse Laplace Transform
2. Convolution

- In practice, the **inverse Laplace transform** is computed by decomposing $H(s)$ into elementary forms for which the Laplace transform is known. The decomposition into elementary form is achieved using partial fraction decomposition. Given rational function $H(s) = \frac{N(s)}{D(s)}$, where $N(s)$ and $D(s)$ are polynomials in s , the goal is to express this function as a sum of simpler terms whose inverse Laplace transforms are easier to compute. The steps involved in computing the inverse Laplace transform are:

1. If the degree of $N(s)$ is greater than or equal to the degree of $D(s)$, perform polynomial division first to reduce the degree of $N(s)$. The remainder can then be decomposed using partial fractions.
2. Factor $D(s)$ into irreducible factors. The denominator may contain linear factors (such as $(s - r_1)^m(s - r_2)$) or quadratic factors (such as $(s^2 + as + b)$).

- **Case 1: Distinct Linear Factors:** If $D(s)$ factors into distinct linear factors $(s - r_1)(s - r_2) \cdots (s - r_n)$, the partial fraction decomposition is:

$$\frac{N(s)}{D(s)} = \frac{A_1}{s - r_1} + \frac{A_2}{s - r_2} + \cdots + \frac{A_n}{s - r_n} \quad (1)$$

where $A_1, A_2, \dots, A_n$ are constants to be determined.

- **Case 2: Repeated Linear Factors :** If $D(s)$ contains repeated linear factors $(s - r_1)^k$, the decomposition takes the form:

$$\frac{N(s)}{D(s)} = \frac{A_1}{s - r_1} + \frac{A_2}{(s - r_1)^2} + \cdots + \frac{A_k}{(s - r_1)^k} \quad (2)$$

- **Case 3: Irreducible Quadratic Factors:** If $D(s)$ contains irreducible quadratic factors $(s^2 + as + b)$, the decomposition takes the form:

$$\frac{N(s)}{D(s)} = \frac{As + B}{s^2 + as + b} \quad (3)$$

where A and B are constants to be determined.

- **Case 4: Repeated Quadratic Factors:** If $D(s)$ contains repeated irreducible quadratic factors $(s^2 + as + b)^k$, the decomposition takes the form:

$$\frac{N(s)}{D(s)} = \frac{A_1s + B_1}{s^2 + as + b} + \frac{A_2s + B_2}{(s^2 + as + b)^2} + \cdots + \frac{A_ks + B_k}{(s^2 + as + b)^k} \quad (4)$$

where $A_1, A_2, \dots, A_k$ and $B_1, B_2, \dots, B_k$ are constants to be determined.

3. Once the constants are determined, write the final partial fraction decomposition. Then, the inverse Laplace transform can be applied to each term of the decomposition.

- (a) The inverse Laplace transform of a term of the form $\frac{A_i}{s-p_i}$ is given by $A_i e^{p_i t}$. When this term represents the impulse response of a system, it can be deduced that the poles, corresponding to p_i , lie in the left half of the complex s -plane if $p_i < 0$. This condition indicates that as time $t \rightarrow \infty$, the impulse response decays to zero. Such behavior implies that the system is stable, since a negative real part of the pole ($p_i < 0$) leads to an exponentially decaying response, signifying that the system's output will eventually fade away over time.
- (b) For a repeated pole of the form $\frac{A_i}{(s-p_i)^n}$, the inverse Laplace transform is given by a term involving both an exponential and a polynomial in t , specifically $A_i \frac{t^{n-1}}{(n-1)!} e^{p_i t}$. In this case, even though the presence of the t^{n-1} term may cause the response to grow initially for small values of t , the exponential factor $e^{p_i t}$ still dominates as $t \rightarrow \infty$. If $p_i < 0$, the exponential decay ensures that the impulse response eventually tends to zero, despite the polynomial factor. This indicates that the system remains stable, as the impulse response still decays to zero over time due to the negative real part of the pole ($p_i < 0$).
- (c) The inverse Laplace transform of a term of the form $\frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ is given by (see scribble for finer details)

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right\} = \begin{cases} \frac{1}{\omega_n} \sin(\omega_n t) u(t) & \text{for } \zeta = 0 \\ \frac{1}{\omega_d} e^{-\zeta\omega_n t} \sin(\omega_d t) u(t) & \text{for } |\zeta| < 1 \\ t e^{-\omega_n t} u(t) & \text{for } \zeta = 1 \\ \frac{1}{\omega_d} e^{-\zeta\omega_n t} \sinh(\omega_d t) u(t) & \text{for } |\zeta| > 1 \end{cases} \quad (5)$$

where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. In this case, the poles of the transfer function lie on the left half of the complex s -plane if $\zeta > 0$. Further, as ζ varies, the poles of the system move along a circle in the complex s -plane with radius ω_n , centered at the origin. For $\zeta = 0$, the poles are purely imaginary and lie on the imaginary axis at $\pm j\omega_n$. As ζ increases from 0 to 1, the poles move counterclockwise along the circle, becoming complex conjugates with negative real parts. At $\zeta = 1$, the poles reach the negative real axis and merge at $s = -\omega_n$. For $\zeta > 1$, the poles become real and distinct, moving further along the negative real axis, and the system becomes overdamped. More importantly, when $\zeta > 0$, the impulse response in time domain approaches 0 as $t \rightarrow \infty$, indicating that the system is stable if $\zeta > 0$.

- It is important to note that the partial fraction decomposition method for obtaining the inverse Laplace transform assumes the system is causal or the signal is one-sided, where the signal exists only for $t \geq 0$. For two-sided signals, however, the region of convergence (ROC) of the Laplace transform plays a crucial role. The ROC not only affects the stability and causality of the system but also determines the precise form of the inverse Laplace transform. Specifically, the ROC provides essential information about the signal's behavior for both positive and negative time values, and thus influences the reconstruction of the full time-domain signal $h(t)$.
- **Finding the system response using the Laplace Transform (Approach II):** Let $h(t)$ denote the impulse response of a system modeled by the differential equation $L[y] = x(t)$. It is important to note that the initial

conditions must be considered when computing the impulse response $h(t)$ of the system. The output $y(t)$ of the system (or the solution of the ODE $L[y] = x(t)$) can be computed using convolution i.e.,

$$y(t) = x(t) * h(t) \quad (6)$$

The output in the s -domain is given by

$$Y(s) = \mathcal{L}\{y(t)\} = \mathcal{L}\{x(t) * h(t)\} \quad (7)$$

This can be further simplified using the following property of Laplace transform:

- **Convolution property:** The Laplace transform of the convolution of input and impulse response is the product of Laplace transform of input and the impulse response. This can be shown as follows:

$$\begin{aligned} \mathcal{L}\{y(t)\} &= \mathcal{L}\left\{\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau\right\} \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau\right] e^{-st} dt \end{aligned} \quad (8)$$

By changing the order of integral,

$$\begin{aligned} \mathcal{L}\{y(t)\} &= \int_{-\infty}^{\infty} x(\tau) \underbrace{\left[\int_{-\infty}^{\infty} h(t-\tau)e^{-st} dt\right]}_{e^{-s\tau}H(s)} d\tau \\ &= H(s) \underbrace{\int_{-\infty}^{\infty} x(\tau)e^{-s\tau} d\tau}_{X(s)} \end{aligned} \quad (9)$$

$$Y(s) = H(s)X(s)$$

The output of the system in s -domain (or the solution of the ODE) is the product of Laplace transform impulse response and the Laplace transform of the input signal. The output in time-domain ($y(t)$) can be computed by taking the inverse Laplace transform of $Y(s)$.

- **Finding impulse response using the Laplace Transform:** The impulse response of the system $h(t)$ can be computed in a simpler manner using the Laplace transform (especially when dealing with ODEs with initial conditions). The Laplace transform allows us to convert the differential equation into an algebraic equation, making it easier to solve for the system's response. The typical steps involved in finding the impulse response are:

- Let $h(t)$ denote the impulse response of the system modeled as $L[y] = x(t)$ i.e., $h(t) = \{y(t) : L[y] = \delta(t)\}$.
- Apply the Laplace Transform to both sides of the differential equation results in an algebraic expression which can be solved to compute the output in s -domain i.e., $H(s)$. When applying the Laplace Transform to the differential equation, it is essential to incorporate the initial conditions.
- After solving for $H(s)$ in the s -domain, the impulse response in time-domain can be obtained by taking the inverse Laplace transform of $H(s)$.